

# DAY-12(03.08.2026)

## **Subnetting:**

divides large IP into smaller ones → subnets → improves network performance, security, communication efficiency, manageability

- Department becomes separate with subnet network where a single network is divided into many networks that limits the broadcast traffic
- Devices can communicate within the department using subnet network
- Network performance → traffic is within the subnet that shared by the devices, security → departments are isolated within the subnet and management → ip used efficiently

## **SUBNET MASK:**

- It is a 32bit → separates network and host IP

## **CIDR notation:**

subnet mask denotes binary digits which is long instead of that CIDR can be used which represents the network part as (/n)

CIDR → Classless Inter Domain Routing

ex: 192.168.10.0/24 → from 32 bits → 24 bits network and 8 bits host

range 0 to 255 → 1st network ID last broadcast IP(256 IP)

ex: 192.168.10.0/26

subnet mask - 11111111.11111111.11111111.11000000

no. of network bits - 26

no. of host bits - 6

no. of host IP address -  $2^6 = 64$  (valid 62 becoz 1st becomes network and last becomes broadcast)

N1 - 192.168.10.0 - 19.168.10.63

N2- 192.168.10.64 - 192.168.10.127

N3- 192.168.10.128 - 192.168.10.191

N4- 192.168.10.192 - 192.168.10.255

**ADVANTAGES:**

- Easy maintenance
- Network traffic is reduced
- Improved Security

**DISADVANTAGES:**

Extra overhead - wastes 2 ip address which denotes as subnet id and broadcast id

High cost → for subnetting additional device is used like routers, switches

More complexity → it adds extra steps for communication